Lab 3: Architecture Justification

Corporate Mental Wellness Platform | Team 14 | PES1UG25AM809 | Section C

Architecture Selection

We chose **Layered Architecture** for the **Corporate Mental Wellness Platform**, an anonymous platform where employees assess stress, book confidential counseling and track wellness. The system is split into a Presentation Layer (Web / Mobile UI), a Business Layer (Authentication & Anonymity, Stress Assessment, Counseling Booking and Wellness Tracker services) and a Data Layer (Encrypted Wellness Database). Layers communicate only through the interfaces IAuth, IAssessment, IBooking, IWellness and IDataAccess.

Two Reasons for the Choice

- **Each feature maps to one clear component.** The three core features (stress assessment, counseling booking, wellness tracking) plus anonymous authentication each become one service in the Business Layer, all following the same flow: UI → service → database. This is easy for a student team to build, test and divide among members. Microservices would add service discovery, network calls and deployment overhead that a platform of this size does not need, and a plain client-server design would mix UI, business rules and data access together.
- **Changes stay contained.** Because every component hides its internals behind an interface, we can change the scoring rules, swap the booking calendar, add a new questionnaire or redesign the UI without rewriting the other layers. Each service can also be tested alone using stubs for the layer below.

Security Advantage

Anonymity is enforced in one place. Only the Authentication & Anonymity Service handles employee identity and issues an anonymous session token, and every other service sees just that token, never a name or employee ID. The UI cannot reach the database directly, so all requests are validated in the Business Layer first, and sensitive data (scores, bookings, wellness logs) is encrypted at rest in the Data Layer. Counselors only see the anonymous alias attached to a booking, which keeps counseling confidential.

Performance Benefit

Layers can be scaled independently. If many employees take assessments at the same time (for example after a company-wide stress survey), only the Business Layer needs more instances behind a load balancer, while the UI and database stay unchanged. Static data such as the questionnaire and available counselor slots can be cached in the Business Layer, so most screens load without a database query.

Trade-off Acknowledged

Passing requests through several layers adds a small delay, and the database is a shared dependency. For an internal company platform, this cost is small compared to the security, maintainability and scalability gains.